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Technical Section

Enhancing semantic mapping in text-to-image diffusion via Gather-and-Bind ☆

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ABSTRACT

Text-to-image synthesis is a challenging task that aims to generate realistic and diverse images from natural language descriptions. However, existing text-to-image diffusion models (e.g., Stable Diffusion) sometimes fail to satisfy the semantic descriptions of the users, especially when the prompts contain multiple concepts or modifiers such as colors. By visualizing the cross-attention maps of the Stable Diffusion model during the denoising process, we find that one of the concepts has a very scattered attention map, which cannot form a whole and gradually gets ignored. Moreover, the attention maps of the modifiers are hard to overlap with the corresponding concepts, resulting in incorrect semantic mapping. To address this issue, we propose a Gather-and-Bind method that intervenes in the cross-attention maps during the denoising process to alleviate the catastrophic forgetting and attribute binding problems without any pre-training. Specifically, we first use information entropy to measure the dispersion degree of the cross-attention maps and construct an information entropy loss to gather these scattered attention maps, which eventually captures all the concepts in the generated output. Furthermore, we construct an attribute binding loss that minimizes the distance between the attention maps of the attributes and their corresponding concepts, which enables the model to establish correct semantic mapping and significantly improves the performance of the baseline model. We conduct extensive experiments on public datasets and demonstrate that our method can better capture the semantic information of the input prompts. Code is available at https://github.com/huan085128/Gather-and-Bind.

1. Introduction

Text-to-image (T2I) generation is a very interesting and challenging field in computer vision. In recent years, with the continuous development of diffusion models, many diffusion-based T2I methods have emerged [1–4], among which the Stable Diffusion based on latent diffusion model [1] has received wide attention. It can generate high-quality and diverse images based on natural language text prompts, showing strong creativity and expressiveness. However, in many cases, the generated images will have elements that do not match or are irrelevant to the text [5–7]. These problems may reduce the user's trust and satisfaction with the generative model and limit its value in practical applications.

Chefer et al. [7] pointed out two key semantic problems that plague the state-of-the-art text-to-image generation models, namely "catastrophic forgetting" and incorrect "attribute binding". This means that the models fail to generate one or more concepts mentioned in the prompt, or assign attributes to the wrong concepts or not fully assign them. To address these issues, they introduced an approach known as "Attend-and-Excite". This training-free method generates cross-attention maps by averaging all 16×16 attention layers and their heads in a single forward pass [8]. In the Unet architecture, these cross-attention layers play a crucial role in determining the spatial layout and geometric shapes of images. Subsequently, the method optimizes the loss objectives during the inference phase, aiming to maximize the peak attention values received by each object token. Agarwal et al. [9] pointed out another key issue, namely the "attention overlap" phenomenon. Specifically, at the inference stage, there is a lot of overlap in the cross-attention corresponding to each concept, which causes two different concepts to be highly activated in the same pixel area. As a result, the final generated image cannot distinguish between these two objects and can only choose one of them.

For instance, in the first row of Fig. 1, under the utilization of stable diffusion, the attention areas of different concepts occasionally overlap. An example of this is the overlapping attention regions of "suitcase" and "elephant", leading to these concepts being indistinct

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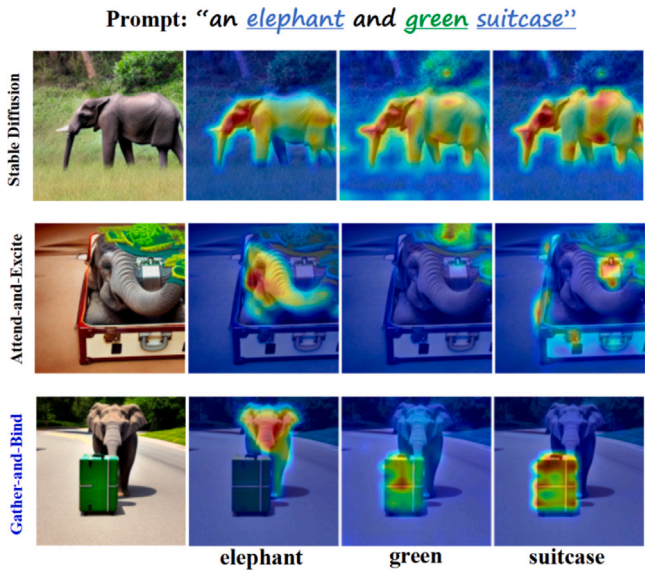


Fig. 1. Compared with the baseline model, our method can better aggregate the cross-attention map to generate a complete concept, rather than multiple incomplete concepts, such as the suitcase in the figure, and can correctly bind the attributes corresponding to the concept.

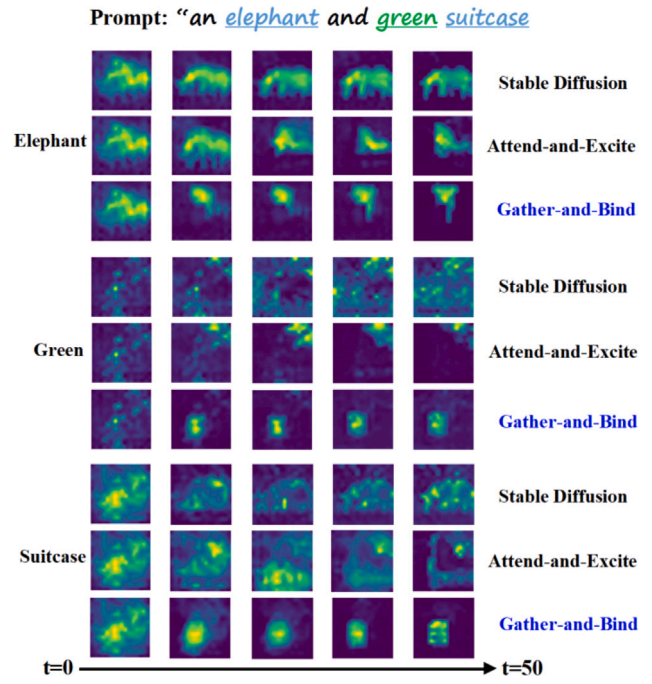


Fig. 2. The change of cross-attention maps with increasing denoising steps. Compared with the baseline model, our method can keep the attention of each concept concentrated and the pixels highly activated in the denoising steps.

in the resulting image. In contrast, the Attend-and-Excite and Gather-and-Bind methods effectively generate images containing the "suitcase" concept. Further experiments indicate that even in the absence of the cross-attention overlap described in Agarwal et al. [9], the issue of excessive attention dispersion of one concept can still arise when the cross-attention maps of two concepts are entirely independent. This dispersion of attention results in generated images that lack clear structure and logic, as illustrated in the second row of Fig. 1. We define this phenomenon as "attention divergence". More on this in the subsequent Appendix A2

In response to this issue, we propose a new training-free method called Gather-and-Bind. This approach enhances the semantic mapping in the text-to-image diffusion process by optimizing two loss functions during the inference stage. Specifically, this method follows the strategy outlined in Refs. [7–9], by calculating the cross-attention maps for each token through a single forward pass. One of the loss functions used is the entropy loss, which treats these cross-attention maps' normalized matrices as probability distributions and uses entropy to measure the dispersion of the cross-attention weights. At the inference stage, we gradually optimize this loss function to make the cross-attention distribution corresponding to each concept more concentrated. As can be seen from Fig. 2, in the process of attention concentration, the pixel areas highly activated by two concepts are also effectively separated. We believe this is because the concentrated attention map retains a distinct highly activated area, which makes the model more easily distinguish between two overlapping pixel areas.

We found another problem with Attend-and-Excite [7], which is that it cannot bind the modifiers (such as "green" in Fig. 2) to the concepts correctly, thus failing to generate a "green suitcase". Our method can effectively achieve the binding of attributes and conceps, making "green" appear in the appropriate position. Rassin et al. [10] analyzed a possible reason why diffusion models generate objects that are inconsistent with the description, namely that they use text encoders such as CLIP [11], which are known to not capture language structure [12]. This makes the diffusion process "lack perception" of language binding, resulting in mismatched results. Therefore, their method is to promote the overlap of the attention map corresponding to the attribute information (such as color, shape, etc.) of the concept with the concept at the inference stage, so that the modifiers can be generated in the correct position. This paper follows their idea, but differs in that we use symmetric Jensen–Shannon-Divergence to measure the distance between two distributions, which performs better than Kullback–Leibler-Divergence in our experiments. Due to space limitations, this will be discussed in the subsequent Appendix B1.

The main contributions of this paper can be summarized as follows:

- we identified two new problems with Attend-and-Excite [7], namely attention divergence and attribute binding error, which lead to generated images that are blurry in structure and logic and semantically mismatched. We propose a new method called Gather-and-Bind, which does not require any training and solely optimizes entropy loss and attribute binding loss during the inference phase to enhance the semantic mapping capabilities of the diffusion model.

2. Related works

Text-to-image generation can be traced back to 2016 with CGANs [13], which used a pre-trained character-level convolutional recurrent neural network to encode text descriptions, and then input text features and random noise into a generator to generate images that match the text. Subsequently, a series of works such as [14–17] appeared. Although these works can generate images with good quality and diversity, they are difficult to match the user's description. With the introduction of diffusion models, text-to-image generation gradually moved towards high controllability and high quality. GLIDE [18] was the first to apply diffusion models to text-to-image generation, and it adopted a classifier-free guidance technique [19], which can generate more delicate and realistic images according to the text description. Latent diffusion [1] generates images by iteratively denoising data in a latent representation space, and then decodes the representation results into complete images, greatly reducing the computational complexity and generating more detailed images. DALL-E2 [2] and Imagen [4] combined the powerful natural language understanding and image representation capabilities of CLIP [11] and T5 [20], and can generate high-quality, diverse and text-relevant images.

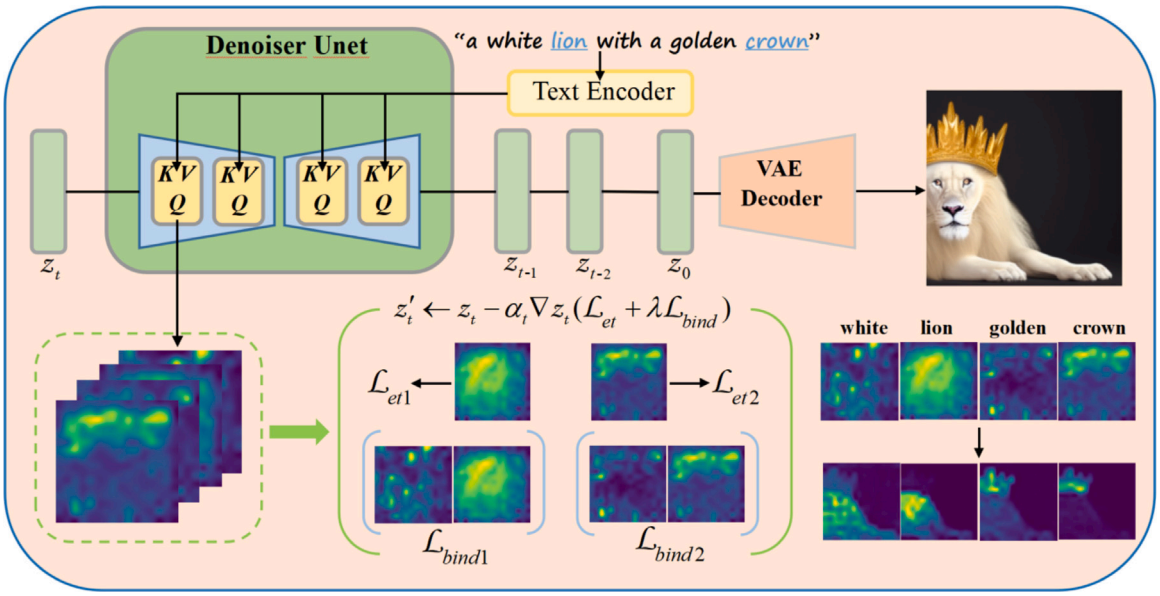


Fig. 3. Overview of the Gather-and-Bind method. In the inference stage, we optimize two loss functions simultaneously, where $\mathcal{L}_{et}$ denotes the information entropy of the noun concepts, which are "lion" and "crown" in the figure, and $\mathcal{L}_{bind}$ represents the distance measure between the nouns and their corresponding modifiers. By minimizing these two loss functions, we make the attention gather and solve the problem of incorrect attribute binding.

Although these text-to-image diffusion models can generate impressive images, they often fail to match the user's prompt, resulting in incorrect generation. Some researchers have proposed solutions based on Generative Adversarial Networks (GANs). For instance, Tan et al. introduced the AMI-GAN model [21], which enhances semantic alignment between text and images through attention-bridged modality interaction and a residual-aware discriminator, thereby generating high-quality images. Yuan et al. [22] proposed the SEMA framework, utilizing adversarial learning of semantic distances to ensure the semantic consistency between generated images and text descriptions. Tan et al. [23] introduced the DR-GAN model, which incorporates semantic decoupling and distribution normalization modules to improve the semantic extraction capabilities of the generator and the distribution learning abilities of the discriminator. However, GAN models face certain limitations regarding training stability and generation diversity.

In text-to-image generation based on diffusion models, one approach to address this issue is to fine-tune the diffusion model for generating customized content [24–27]. Although these methods allow diffusion models to generate images that meet user requirements, they necessitate additional datasets and extended fine-tuning periods. One way to solve this problem is to fine-tune the diffusion model to generate customized content [24–27], although these methods can make the diffusion model generate images that meet the user's needs, they require additional datasets and long fine-tuning. Hertz et al. [8] found that the spatial layout and geometric shape of the image are determined by the cross-attention layer in Unet, rather than by the text encoder or other parts. As a result, many researchers turned their attention to controlling the attention map [5–7,9,10,28–30], by interfering with the cross-attention map at the inference stage to achieve the effect of controlling the model generation. Feng et al. [5] incorporated language structure into the structured diffusion guidance of the cross-attention layer to more accurately bind attributes, while maintaining the overall image quality and diversity. Chefer et al. [7] maximized the activation of the most neglected concept in the generated output to manipulate the cross-attention map to solve the concept omission problem, but failed in some cases, as shown in Fig. 7. Agarwal et al. [9] proposed that the diffusion model has two problems of attention overlap and attention decay, and they designed an attention separation and preservation loss to solve these two problems. We take a different perspective and re-examine the cross-attention map corresponding to each concept at inference time, and find another problem, which we call attention divergence, which can still cause attention divergence even if the attention does not overlap, resulting in failure to correctly match semantic information. Therefore, we designed a novel information entropy loss to solve this problem. Rassin et al. [10] focused on solving the attribute binding problem, they proposed using average Kullback Leibler Divergence to measure the distance between attributes and concepts, and minimized this distance at inference time to match semantic information. Our attribute binding loss is based on this idea.

3. Methodology

3.1. Preliminary

Stable Diffusion Model. We implemented our method and experiments on the state-of-the-art T2I model "Stable Diffusion" [1]. First, the Stable diffusion uses a pre-trained autoencoder [31] $\varepsilon(\cdot)$ and $D(\cdot)$ to map the input image x to a low-dimensional latent space. The encoder $\varepsilon(\cdot)$ transforms the image x into a latent code z, where $z = \varepsilon(x)$, and the decoder $D(\cdot)$ is to reconstruct the input image from the code z in the latent space, i.e. $D(\varepsilon(x)) \approx x$. This pre-training stage of the autoencoder aims to learn how to capture the key features of the image. In the denoising process, by random sampling of noise $\epsilon \sim \mathcal{N}(0, I)$ and time steps t, we can obtain the noisy latent code $z_t = \alpha_t z + \sigma_t$, where α_t and σ_t are coefficients that control the noise schedule. Then, DDPM [32] is conditioned on an additional input vector, usually a text encoding generated by a pre-trained CLIP text encoder. Specifically, the DDPM model is trained according to the following denoising objective:

$$\mathcal{L} = \mathbb{R}_{z\sim\varepsilon(x),y,\epsilon\sim\mathcal{N}(0,I),t}\left[\|\epsilon - \epsilon_\theta(z_t, t, \tau_\theta(y))\|_2^2\right], \tag{1}$$

This process involves generating and denoising noisy latent code z_t, where the denoising network ϵ_θ is to accurately remove noise, given noisy latent z_t, time step t and conditional $\tau_\theta(y)$ encoding. In the inference process, random Gaussian noise z_t is iteratively denoised to, and the final image is obtained through the decoder $x' = D(z_0)$.

Cross Attention. The Stable diffusion introduces a cross-attention mechanism, which allows the model to adjust the spatial layout and content of the image according to the given text prompt during the image generation process. Specifically, the input text prompt W =

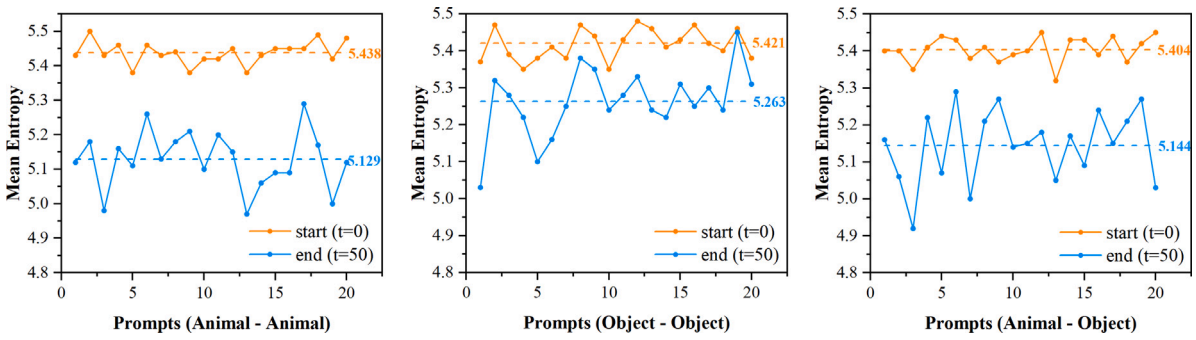


Fig. 4. This figure illustrates the average maximum information entropy of the Stable Diffusion model at$t$ = 0 and t=50. The horizontal axis represents different prompts, with each category consisting of 20 distinct prompts. The dashed line represents the average of these values.

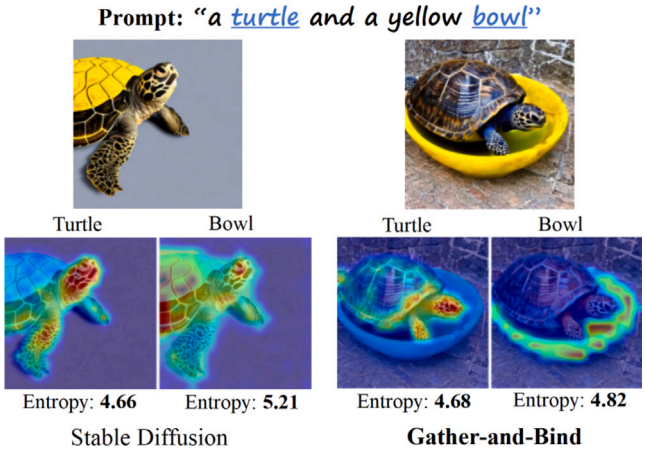


Fig. 5. The visual results of cross-attention maps. When the information entropy of a particular concept becomes excessively high, Stable Diffusion often tends to neglect it.

$(w_1, w_2, \ldots, w_n)$ is transformed into text embedding $c = f_{CLIP}(W) \in \mathbb{R}^{n \times e}$ features by a pre-trained text encoder (such as CLIP), where e is the embedding dimension. These embedding features are passed through two linear projection layers, resulting in corresponding *key* $K \in \mathbb{R}^{n \times d}$ and *value* $V \in \mathbb{R}^{n \times d}$. At the same time, the local features of the image are also passed through a linear projection layer, resulting in corresponding query vectors *query* $Q \in \mathbb{R}^{n_q \times d}$, and then the attention map is obtained by calculating the dot product between the query vectors and the key vectors:

$$M = Softmax(\frac{QK^T}{\sqrt{d}}), \tag{2}$$

where M_{ij} denotes the weight of the jth token value at pixel i, d is the latent projection dimension of keys and queries. Finally, the cross-attention output is defined as:

$$Attention(Q, K, V) = Softmax(\frac{QK^T}{\sqrt{d}})V, \tag{3}$$

3.2. Calculating information entropy

As stated in Section 1, we found another problem: the phenomenon of attention dispersion. Fig. 2 illustrates that for the"suitcase"concept, the cross-attention of Stable Diffusion remains scattered and these attention weights decay as the denoising step progresses, leading to the attention overlap problem reported by Agarwal et al. [9]. Attend-and-Excite [7] aims to guide the model to attend to all concept tokens and amplify their activation during denoising step, but solely concentrating on the largest attention weight will inevitably overlook others that are relatively smaller, resulting in a final image that is messy and disorganized. Our approach is to perturb the attention map during denoising to make it gradually aggregate into a highly activated pixel region that can form rapidly in the early stage of denoising and persist in the later stages.

We consider information entropy to measure the degree of concentration of each cross-attention map. Specifically, given the noisy latent z_t the current time step t, we perform forward propagation through the pre-trained UNet network using z_t and text prompt $\mathcal{P}$. Then, we consider the cross-attention map obtained after averaging over all 16×16 attention layers and heads. The generated aggregated map A_t contains N cross-attention maps, one for each token of $\mathcal{P}$, i.e., $A_t \in \mathbb{R}^{16 \times 16 \times N}$. We perform a normalization operation on each cross-attention map, so that the sum of all elements inside it is 1. After the normalization operation, the (i, j)th element of the resulting matrix A_t represents the probability of each text token appearing in the corresponding image block, denoted as $\mathcal{P}(A_{ij})$, so that we can approximate this attention matrix as a random probability distribution. Then, we calculate the information entropy for each attention map corresponding to a concept:

$$H(A_t) = -\sum_{i,j} p([A_t]_{ij}) \log p([A_t]_{ij}), \tag{4}$$

For a 256-dimensional random variable, at this time $H(A_t) \in [0, 8]$, when it is uniformly distributed, it gets the maximum value. The smaller the value of $H(A_t)$, the more concentrated the attention weight distribution is, and the larger it is, the more dispersed it is. We randomly sampled 20 prompt words from each subset of the Attend-and-Excite [7] dataset, and used 64 random seeds for each prompt word, then performed inference using Stable Diffusion to generate images. In this process, we calculated the average information entropy at the denoising steps t = 0 and t = 50. For example, for the prompt word "a turtle and a yellow bowl", we calculated the information entropy of the cross-attention maps corresponding to turtle and bowl at t = 0, and took the maximum value among them, then averaged with the results of all seeds. Similarly, we did the same calculation at t = 50. Our final results are shown in Fig. 4. From the results, we can see that at the initial t=0, the maximum information entropy is around 5.4 for all cases, and it slightly decreases at t = 50. At t = 0, the latent image is merely a 64×64 pure noise image, and the 16×16 cross-attention map appears disordered and chaotic, leading to a high mean entropy initially. As the denoising process progresses, the cross-attention map gradually concentrates and forms complete patterns, which indicate the contours of the final generated image, such as the lion and crown shown in Fig. 3. However, the Stable Diffusion model often fails to fully match descriptions when prompted with multiple concepts, typically generating only some of the concepts. In such cases, the entropy of the ignored concepts in the cross-attention map is usually higher, while the concepts that are successfully generated show lower entropy, as illustrated with "bowl" and "turtle" in Fig. 5.

Table 1
Escription of benchmarks used for the experimental evaluation.

Evaluation Set	Description	Example
Animal-Animal	a [animalA] and a [animalB]	"a cat and a frog"
Animal-Object	a [animalA] and a [colorB][objectB]	"a turtle and a pink balloon"
Object-Object	a [colorA][objectA] and a [colorB][objectB]	"a pink crown and a purple bow"
COCO-Animal	COCO caption containing animal related from TIFA	"Cat perched on the sofa looking out of the window
COCO-Color	COCO caption containing color related from TIFA	"A white plate topped with broccoli and a dumpling"
COCO-object	COCO caption containing object related from TIFA	"A stop sign sitting on top of a metal column"
COCO-2017val	COCO 2017 validation caption	"Off white toilet with a faucet and controls"

Entropy reflects the concentration of the cross-attention map. When the attention is more focused, the entropy is lower, indicating that certain areas are highly activated. However, if the entropy becomes too low, the attention may overly concentrate on a very small area, potentially causing the generated concept to appear distorted in proportion or form, thus affecting the overall quality of the image. Conversely, higher entropy indicates dispersed attention, which can lead to catastrophic forgetting or overlapping attention issues during the denoising process. For example, the overlap of attention between "bowl" and "turtle" in Fig. 5 resulted in the model only generating one of the concepts or a mix of both, such as incorrectly generating a "yellow bowl" on the turtle shell.

Our proposed method effectively guides the concentration of the attention map and maintains its activation during the denoising process. For instance, when the entropy of "bowl" decreases from 5.21 to 4.82, the cross-attention maps of the two concepts are well-preserved and separated, enabling the model to successfully generate the corresponding images. Therefore, we believe that at t=50, the entropy remains relatively high, indicating that the cross-attention map is still relatively dispersed. Under these conditions, the model is prone to ignoring relevant concepts during the denoising process and may also cause issues with overlapping attention.

3.3. Information entropy loss

As mentioned above, we analyzed the reason for the catastrophic neglect of Stable diffusion from the degree of concentration of the cross-attention maps, which Attend-and-Excite [7] did not solve. Our key insight to solve this problem is simple: guide the model to pay attention to multiple concepts simultaneously by assisting the diffusion model to aggregate the attention maps. By doing so, we force the DDPM denoising process to aggregate the attention maps of each concept, forming separate, highly activated regions. This ultimately leads to z_0 representing all the concepts that can be decoded into the desired image. We implement this idea through information entropy loss. Therefore, for an input prompt containing multiple noun concepts $\{N_1, N_2, N_3, \ldots, N_k\}$, each N_i representing a noun concept, based on Eq. (4), the information entropy loss corresponding to each concept can be defined as:

$$\mathcal{L}_{et\{N_i\}} = H(A_t^{\{N_i\}}), \tag{5}$$

The total loss is then taken as the maximum value among them:

$$\mathcal{L}_{et} = max(\mathcal{L}_{et\{N_1\}}, \mathcal{L}_{et\{N_2\}}, \ldots, \mathcal{L}_{et\{N_k\}}), \tag{6}$$

The core idea is to concentrate the weight distribution and separate multiple highly activated regions by calculating and reducing the maximum information entropy of the cross-attention maps.

3.4. Attribute binding loss

As discussed in Figs. 1 and 2, Attend-and-Excite [7] actually did not solve the problem of incorrect attribute binding, although it can generate the attribute"green", whereas the generated region does not overlap with the concept. Therefore, we propose to use attribute binding loss to force the concept to match its corresponding attribute, which is mainly inspired by the work of [10]. In this paper, we use the symmetric and bounded Jensen–Shannon-Divergence (JSD) as the distance measure. JSD is an extension of KL divergence, which solves the problem of non-symmetry of KL divergence, and it performs better in our experiments. Considering a pair of noun phrases (m,n)(m,n), such as ("white", "lion"), our goal is to use JSD to measure the degree of overlap between the two attention maps, so the loss function can be defined as:

$$\mathcal{L}_{bind} = \frac{1}{2} D_{KL}(A_m \parallel \frac{A_m + A_n}{2}) + \frac{1}{2} D_{KL}(A_n \parallel \frac{A_m + A_n}{2}), \tag{7}$$

where A_m and A_n represent the normalized cross-attention maps corresponding to a pair of noun phrases (m, n), respectively. The KL divergence is defined as:

$$D_{KL}(A_i \parallel A_j) = \sum_{pixels} A_i log(A_i / A_j), \tag{8}$$

We force the diffusion model to minimize this measure as much as possible during the denoising process, so that the cross-attention map of the modifier overlaps largely with the cross-attention map of the noun.

3.5. Optimization flow

Our entire optimization process is shown in Fig. 3. First, we extract the 16 × 16 cross-attention maps corresponding to each noun concept and noun phrase. Then we calculate the information entropy loss $\mathcal{L}_{et}$ and attribute binding loss $\mathcal{L}_{bind}$ separately, and the total loss is:

$$\mathcal{L} = \mathcal{L}_{et} + \lambda \mathcal{L}_{bind}, \tag{9}$$

Since the information entropy loss is much larger than the attribute binding loss, we set λ to 3 in our experiments. After calculating the loss, we update the latent code using gradient descent:

$$z_t' \leftarrow z_t - \alpha_t \nabla_{z_t} \mathcal{L}, \tag{10}$$

where α_t is a scalar that defines the gradient update step size, which we set to $t = T$. We only perform updates from 0 to $t = tend$, where $T = 50$, $tend = 25$. For token recognition, we use spaCy's transformer-based dependency parser [33] to parse the prompt, and automatically identify the noun concepts and noun phrases in it.

4. Results

4.1. Benchmarks

We conducted a detailed assessment of the seven prompt sets summarized in Table 1, which included evaluations of CLIP text–image similarity, BLIP text–text similarity, as well as TIFA [34] and FID assessments. A&E [7] designed three types of prompts: (i) "a [animalA] and a [animalB]", (ii) "a [animal] and a [color][object]", and (iii) "a [colorA][objectA] and a [colorB][objectB]". From these, they created 276 textual prompts, some of which we selected for qualitative comparison, although our study is not limited to these examples. To better demonstrate the advantages of our approach, we also designed more complex prompts as shown in Fig. 6.

Additionally, we introduced the TIFA (Text-to-Image Fidelity Assessment with Q&A) benchmark, an automated evaluation tool that



Fig. 6. Qualitative comparison of Gather-and-Bind with other state-of-the-art methods. (1) The first and fourth columns are from the dataset [7], and the second and third columns are prompts that we construct ourselves. (2) For each prompt, we generate four images using each method and show the best one among them.

Table 2
Average CLIP text–text similarities between the text prompts and captions generated by BLIP over the generated images.

Method	Animal - Animal	Animal - Object	Object - Object
StableDiffusion [1]	0.761 (−7.10%)	0.785 (−6.75%)	0.769 (−4.55%)
ComposableDiffusion [6]	0.689 (−18.29%)	0.772 (−8.55%)	0.760 (−5.79%)
StructureDiffusion [5]	0.763 (−6.81%)	0.778 (−7.71%)	0.755 (−6.49%)
Attend-and-Excite [7]	0.804 (−1.37%)	0.825 (−1.58%)	0.808 (+0.49%)
Gather-and-Bind	0.815	0.838	0.804

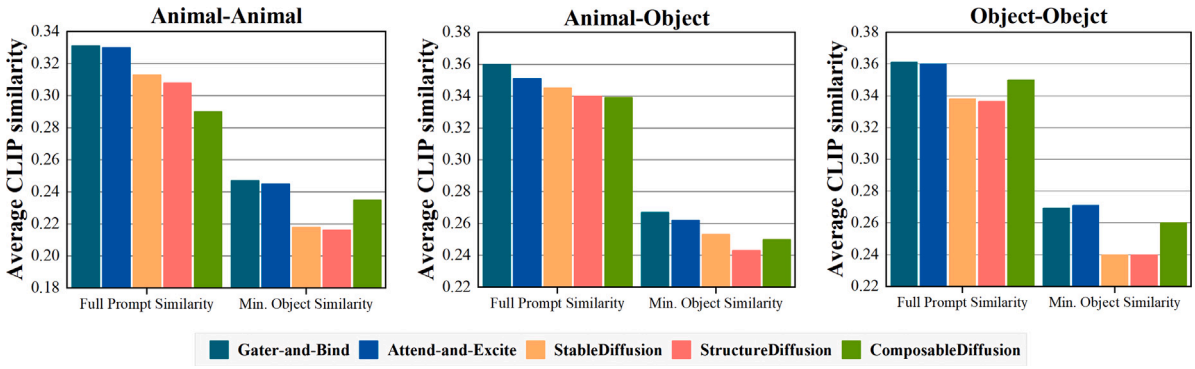


Fig. 7. Average CLIP image–text similarities between the text prompts and the images generated by each method.

Table 3
Detailed evaluation of each text-to-image model on TIFA v1.0.

Method	COCO-Animal	COCO-Color	COCO-Object
StableDiffusion [1]	0.809 (−4.66%)	0.785 (−6.61%)	0.775 (−4.77%)
Attend-and-Excite [7]	0.838 (−2.73%)	0.807 (−1.25%)	0.802 (−1.25%)
Gather-and-Bind	0.844	0.829	0.812

Table 4
Quantitative comparison on the COCO validation set.

Method	FID
StableDiffusion [1]	38.91
Attend-and-Excite [7]	36.09
Gather-and-Bind	36.52

measures the fidelity of generated images to their textual inputs through Visual Question Answering (VQA). Specifically, for each text input, several question–answer pairs are automatically generated using a language model. We then assess image fidelity by examining whether existing VQA models can accurately answer these questions using the generated images. In this study, we utilized TIFA v1.0, which consists of a benchmark with 4000 different text inputs and 25,000 questions across 12 categories (such as objects, counting, etc.). We evaluated our proposed method using TIFA v1.0, although we did not test all categories, choosing only the three types related to animals, colors, and objects, as other categories like material or spatial do not suit our assessment, and neither our method nor other comparative methods were effective in these areas. Finally, we also assessed the quality of the generated images using the FID metric based on the COCO 2017 validation set.

4.2. Qualitative results

As shown in Fig. 6, we compare our method with previous works (including Composable Diffusion [6], Structure Diffusion [5], and Attend-and-Excite [7]). It can be clearly seen from the figure that when dealing with prompts involving multiple concepts, Stable Diffusion, Composable Diffusion, and Structure Diffusion all suffer from serious concept omission problems. Composable Diffusion seems to mix two different concepts together, such as a dog-shaped car in the second column (bottom right image), and a cat-frog hybrid in the third column.

Structure Diffusion generates images similar to Stable Diffusion, both of which fail to capture the semantic information in the text prompts, and also cannot associate colors with concepts well, resulting in a white car and a green bench. In our experiments, Attend-and-Excite can handle the concept omission problem better, but still has two shortcomings. First, the generated features are chaotic, such as the cat with frog claws in the third column (the last two images), the backpack-bench fusion in the fourth column (bottom left image), or multiple connected backpack-like objects (bottom right image). Second, the attribute binding is incorrect, such as generating red apples and blue cars. Our Gather-and-Bind method solves these two problems well by aggregating the attention maps of each concept in the denoising process and overlapping the attention maps of attributes and concepts, successfully improving the quality of the generated results.

4.3. CLIP and BLIP analysis

We follow the evaluation method of [7] to quantify the performance. Specifically, we use random seeds to generate 64 images for each prompt in the dataset, and then use CLIP to compute the average cosine similarity between the generated images and the text. This is the full prompt similarity. However, a study [35] pointed out that CLIP exhibits some instability when computing similarity, which mainly stems from its selective similarity scoring mechanism that only considers a subset of semantically meaningful input tokens. To address this issue, we introduce the minimum object similarity. First, we split the input text prompt into two parts by conjunctions (such as “and”, “with”), and then use CLIP to separately compute the cosine similarity between the generated image and the split prompt tokens. Finally, we take the smaller score of the two. We average the scores for all prompts and obtain the results shown in Fig. 7. From the figure, we can observe that Gather-and-Bind achieves comparable results on the Animal-Animal and Object-Object subsets. In particular, it performs excellently on the Animal-Object subset, improving by 2.5% and 1.9% over Attend-and-Excite (and by 4.3% and 5.5% over Stable Diffusion). However, on the Object-Object subset, the minimum object similarity is slightly lower than Attend-and-Excite. This is due to some cases in this subset where there are prompts with different concepts but same colors, such as “a white suitcase and a white chair”. In this case, our attribute binding method may have a negative effect. Specifically, the goal of the attribute binding loss is to make the two attention maps overlap as much as possible. When there are multiple same attributes in one prompt token, it will cause the corresponding concepts to also overlap as much as possible, resulting in a generated image that may fuse the features of both concepts (see Appendix B2 for details).

Further, we use a pre-trained BLIP [36] image captioning model to add captions on the generated images, and then compute the average CLIP similarity between the prompt tokens and the captions. The specific results are shown in Table 2. As shown in the table, the Gather-and-Bind method performs better than Attend-and-Excite on the Animal-Animal and Animal-Object subsets, which clearly demonstrates the effectiveness of our method. However, on the Object-Object subset, our method does not achieve the best result, slightly lagging behind Attend-and-Excite, which can be traced back to the adverse effect of the attribute binding loss on the performance mentioned earlier.

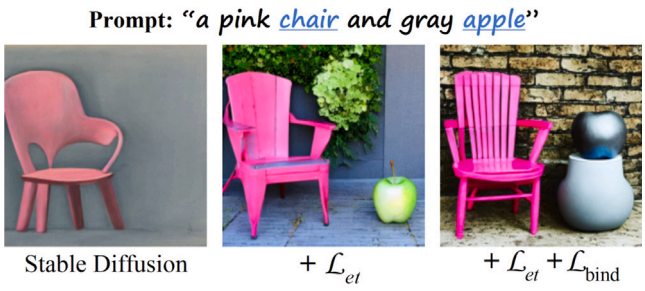


Fig. 8. Ablation study under the same random seed.

Table 5
Ablation results for text–text similarities.

Method	Animal - Animal	Animal - Object	Object - Object
SD	0.761	0.785	0.769
SD+$\mathcal{L}_{et}$	0.815 (+7.10%)	0.828 (+5.48%)	0.810 (+5.33%)
SD+$\mathcal{L}_{et}$+$\mathcal{L}_{bind}$	0.815 (+7.10%)	0.838 (+6.75%)	0.804 (+4.55%)

4.4. TIFA and FID analysis

However, due to CLIP's subpar performance in computing objects [11] or combinational reasoning However, due to CLIP's subpar performance in computing objects [37], the reliability and accuracy of CLIPScore are often questioned. Another set of evaluation metrics involves using image captions where an image captioning model first converts images into text, then assesses these captions by comparing them with the original text input. Unfortunately, relying solely on caption models is insufficient, as they may choose to overlook significant information in the images or focus on irrelevant areas [38], and evaluating the text (caption) generation itself poses challenges [39, 40]. Consequently, we also employed the TIFA benchmark, a new assessment tool, to further evaluate the effectiveness of our method.

Table 3 shows the performance of Stable Diffusion, Attend-and-Excite, and our Gather-and-Excite method on TIFA v1.0. According to the results, our method consistently outperformed the others in the three categories of prompts tested, with the most significant improvements observed in queries related to color. This success is attributed to the attribute binding loss in Gather-and-Bind, which effectively ties colors to the correct subjects, ensuring they appear in the right locations, such as in the example "a green backpack and a purple bench" shown in Fig. 6. In contrast, Attend-and-Excite often mistakenly produces a green bench or a purple backpack, whereas Gather-and-Bind accurately generates images that match the descriptions.

Additionally, we evaluated the quality of the generated images more specifically. We assessed the FID scores based on the COCO 2017 validation set, where FID is an index measuring the quality of generated images; a lower FID indicates a closer resemblance to real images and, consequently, higher quality. This dataset includes 5000 captioned images. The results, as shown in Table 4, indicate that Attend-and-Excite had the lowest FID score of 36.09 among the three, suggesting that this model produces images of quality closest to real images. Gather-and-Bind's FID score was 36.52, also very close to the performance of Attend-and-Excite. Thus, when considering the results from TIFA v1.0, Gather-and-Bind overall outperforms Attend-and-Excite in terms of performance.

4.5. Ablation study

In Fig. 8, we present the results of gradually introducing different losses on top of the baseline Stable Diffusion. After adding the information entropy loss, we gradually observe that the cross-attention maps of the chair and the apple converge and separate during the denoising process, enabling the model to eventually generate an apple. However, at this point, the gray color is not correctly bound to the apple, but instead generated on the chair. Therefore, when we further introduce the attribute binding loss, the gray attention map overlaps with the apple, resulting in a gray apple being generated. We also quantify the effect of each loss in Table 5, where we use BLIP to generate captions for each image, and compute the average text similarity. As shown in Table 5, after introducing the information entropy loss, the model performance improves significantly on all three subsets, increasing by 7.10%, 5.48%, and 5.33%, respectively. After introducing the attribute binding loss, the Animal-Object subset with color attributes further improves, while the Animal-Animal subset with only concepts remains unchanged. As we mentioned in the quantitative analysis, the attribute binding loss may cause the model to have difficulty distinguishing cases with different concepts but same colors, so keeping only the information entropy loss may yield better results.



Fig. 9. Limitations. Occasional concept blending due to the limited expressive capacity of Stable Diffusion.

5. Limitations

Despite the enhancement of semantic mapping in our approach during text-to-image diffusion, resulting in generated images that align better with the prompt phrases, there are still some limitations to consider. These limitations arise from the finite expressive capacity of diffusion models and the Gather-and-Bind method, which has not undergone additional training. Consequently, in certain cases, the model may struggle to effectively differentiate between some concepts, leading to the blending of distinct concepts, as illustrated in Fig. 9, where the left image confuses the elephant's trunk and tusks with the bear's head, and the right image often generates crowns on eyeglass frames or lenses. Furthermore, in our experiments, we also observed that the Stable Diffusion model cannot precisely control the number and placement of generated concepts, resulting in occasional generation of unrealistic or partially missing concepts. Therefore, achieving precise semantic generation still requires further research and improvement, presenting a challenging task.

6. Conclusions

In this work, we adopt a different approach to analyze the problem of stable diffusion models failing to capture all concepts correctly. By visualizing the cross-attention maps of the concepts during the denoising process, we find that the attention maps of those concepts that are easily ignored are usually quite scattered. Further computing the information entropy, we find that these concepts have much higher information entropy than those that are normally generated. Therefore, we propose an inference-time intervention method called Gather-and-Bind, which does not require any fine-tuning. This method introduces two new loss functions, namely information entropy loss and attribute binding loss. The information entropy loss aims to gradually gather the attention maps of the concepts during the inference process, forming a highly activated pixel region. This region can be well preserved as the denoising steps increase. On the other hand, the attribute binding loss ensures that attributes such as color overlap correctly with the concepts. The joint effect of these two loss functions largely improves the semantic issues of stable diffusion models. Finally, we conduct extensive qualitative experiments on public datasets, demonstrating that our method better conforms to the input text prompts than state-of-the-art methods.

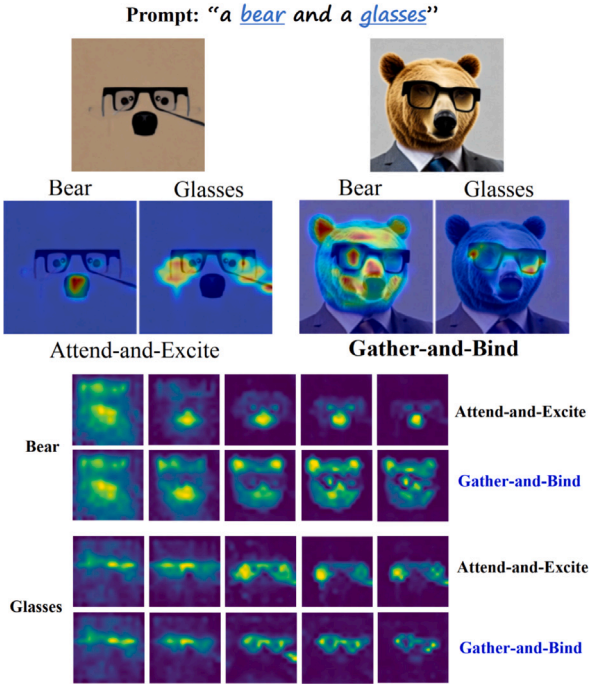


Fig. 10. Visualize Attend-and-Excite and Gather-and-Bind cross-attention with the same random seed.

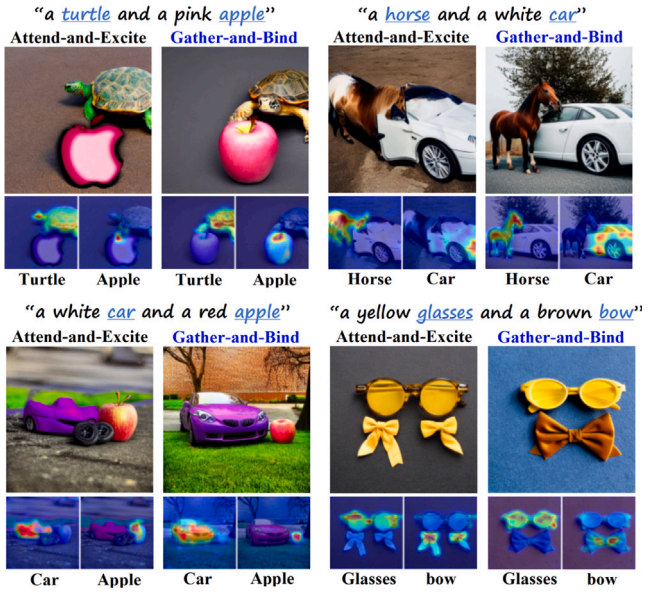


Fig. 11. More cross-attention visualizations with the same random seed.

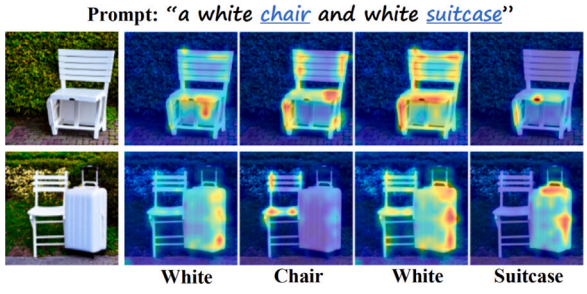


Fig. 12. Cross-Attention Comparison with and without Attribute Binding Loss. The first row depicts with the loss, while the second row does not.

Table 6
Results for different values of entropy difference.

Entropy	Animal - Animal	Animal - Object	Object - Object
Minor Gap	0.233	0.243	0.258
Large Gap	0.276 (+18.5%)	0.313 (+28.8%)	0.291 (+12.7%)

CRediT authorship contribution statement

Huan Fu: Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Guoqing Cheng:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization.

Declaration of competing interest

We declare that there is no conflict of interest regarding the publication of this paper.

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Appendix A

A1. Experimental detail

In all our experiments, we employed the official Stable Diffusion v1.5 [1] text-to-image model, utilizing NVIDIA GeForce RTX 3090 hardware. We updated the latent code using Eq. (10), with a learning rate that followed a linear decay schedule, gradually decreasing from 20 to 10 during the updating process. Refinement iterations were executed when the loss exceeded a certain threshold. Throughout the refinement iterations, the latent code was continuously updated to align with the desired objectives, with a maximum update limit of 20 iterations. We set the loss thresholds as 0: 5.4, 10: 5.2, 20: 4.8, signifying our intention to reduce the loss to below 5.4 at the outset of denoising, reaching below 5.2 by the 10th step, and ultimately descending below 4.8 by the 20th step. The total denoising steps in our experimental setup amounted to 50, with 25 update steps performed.

A2. Attention divergence

As mentioned earlier, the issue of concept omission in diffusion models during the inference phase primarily stems from the phenomenon of excessive attention divergence in the cross-attention maps. In Section Three, we explored the information entropy changes during the denoising process in the Stable Diffusion model. The results in the main text reveal that the Stable Diffusion model maintains a relatively high level of information entropy even after denoising is completed. To further validate our standpoint, we calculated the differences in information entropy for each concept before and after denoising. We segmented the prompt phrases into two parts using conjunctions such as "and" and "with", for instance, splitting "a cat and a dog" into "a cat" and "a dog", then computed the CLIP scores between the generated images and the segmented prompt words for each prompt. We categorized the information entropy differences for each concept in each prompt into "Minor Gap" and "Large Gap" and subsequently calculated the averages for all prompts within each subset. The final results are

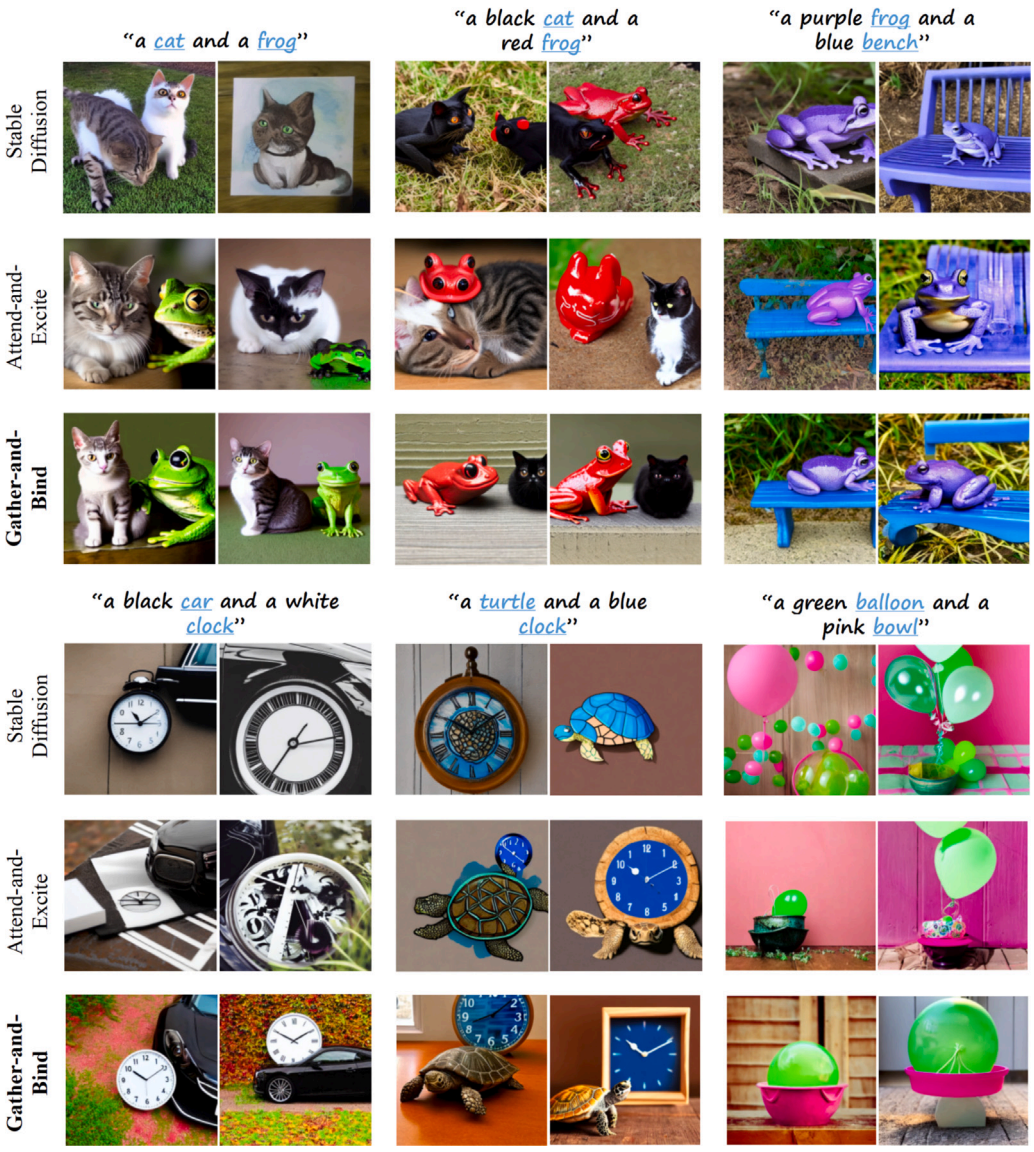


Fig. 13. More qualitative comparison results under the same random seed.

presented in Table 6. It is observed that concepts in prompts with larger information entropy differences exhibit a stronger alignment with the generated images, showing an average improvement of 20% across these three subsets. This suggests that reducing information entropy contributes to mitigating catastrophic forgetting issues.

A3. Additional cross-attention map visualizations

We present further visual results in Figs. 10 and 11. Fig. 10 reveals instances where the Attend-and-Excite [7] failed to capture complete concepts. This incompleteness is not attributed to attention overlap phenomena [9] but rather stems from the core idea of the Attend-and-Excite model, which activates a single attention with the maximum value. This approach may result in attention maps being overly concentrated in local regions. For example, in Fig. 10, the attention map corresponding to "bear" overly focuses on the nose, ultimately leading to incomplete concept generation. In contrast, the Gather-and-Bind method guides the model to aggregate multiple attention maps, not solely relying on a single maximum attention value. Fig. 11 further illustrates the aforementioned phenomenon. In the top-left image, the model fails to generate a complete "apple", in the top-right and bottom-left images, it cannot produce a complete "car", and in the bottom-right image, the model generates multiple "bow" instances with incorrect colors.

Appendix B

B1. KL divergence and JS divergence

As outlined in main text, we conducted experimental comparisons using JS divergence and performed quantitative testing with two different attribute binding losses. By computing the average CLIP text–text similarity, we obtained results as shown in Table 7. Relative to KL divergence, JS divergence exhibited superior performance in the Object-Object and Animal-Object subsets, with improvements of 1.09% and 1.77%, respectively. This is attributed to the slight differences in the generated images produced by both methods under the same random seed. In general, the quality of images generated by JS divergence tends to be higher.

B2. Shortcomings of attribute binding loss

We have observed that in certain cases, when the prompt contains multiple distinct concepts with the same color attribute, the generated images may become confused. As depicted in Fig. 12, there are two instances of "white" in the prompt, but they correspond to different

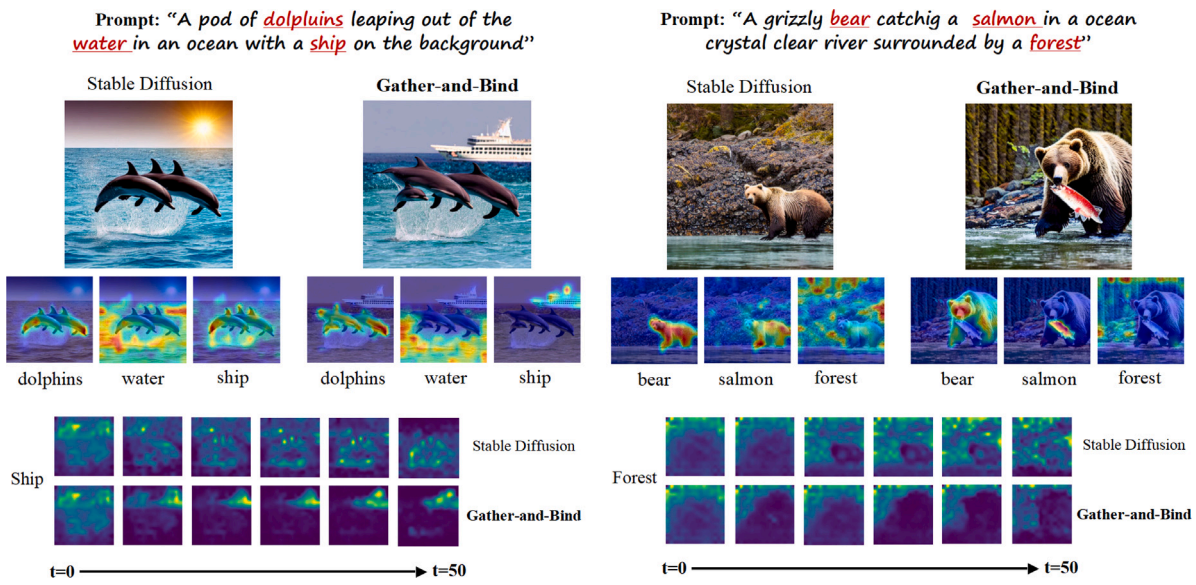


Fig. 14. Effectiveness of our method on prompts containing three concepts.

Table 7
Average CLIP text–text similarities between the text prompts and captions generated by BLIP over the generated images.

Entropy	Animal - Animal	Animal - Object	Object - Object
KLD	0.815	0.829	0.790
JSD	0.815	0.838 (+1.09%)	0.804 (+1.77%)

concepts. The mechanism of the attribute binding loss causes overlapping attention maps related to color associations with different concepts, leading to confusion. In such instances, applying only the information entropy loss can effectively generate two distinct concepts without causing confusion.

B3. More qualitative results

In Fig. 13, we present additional qualitative results. For simple prompts such as "a cat and a frog", Attend-and-Excite typically aligns well with the semantic description. However, it often exhibits suboptimal performance in more complex scenarios. For instance, in the third column, the colors of the "bench" and "bowl" do not align with the textual description, and in the second column, the "turtle" and "clock" are combined, while Gather-and-Bind tends to adhere more closely to the semantic description in such cases.

In Fig. 14, we demonstrate the effectiveness of our method on prompts containing three concepts. It is evident that, compared to the baseline model, our method successfully retains concepts such as "ship", "salmon", and "forest", which are often overlooked by diffusion models.

Data availability

Data will be made available on request.

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